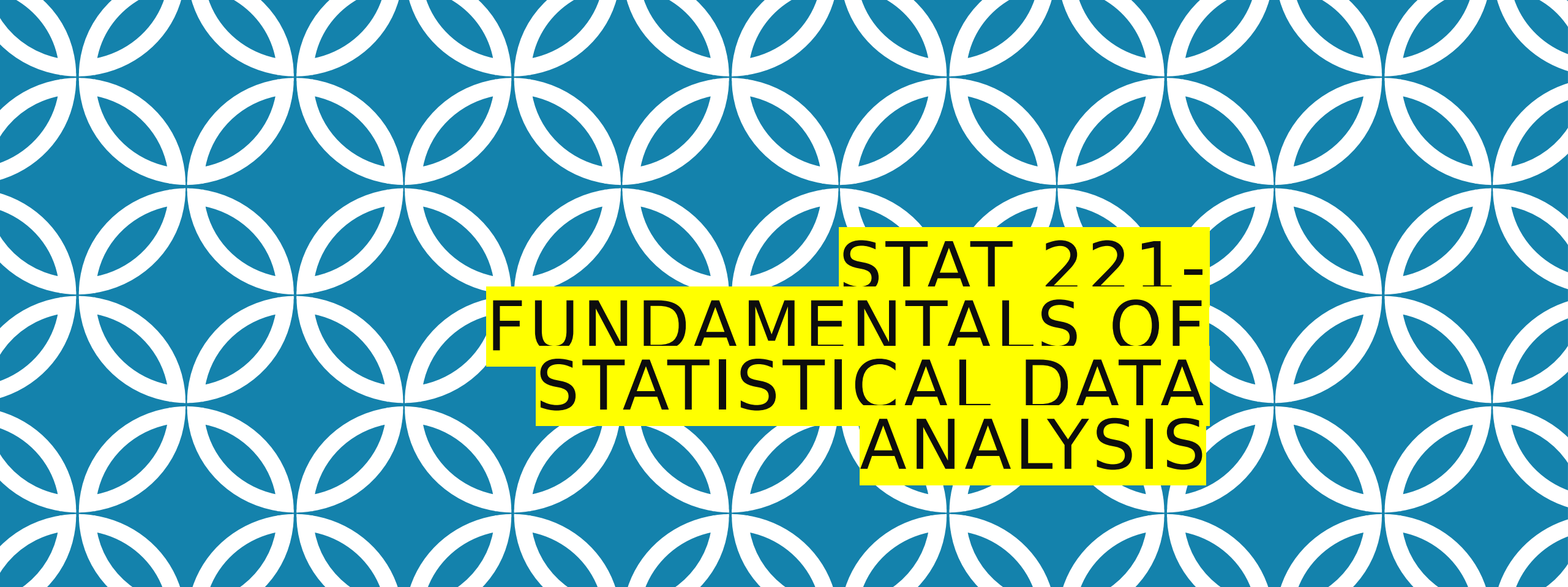
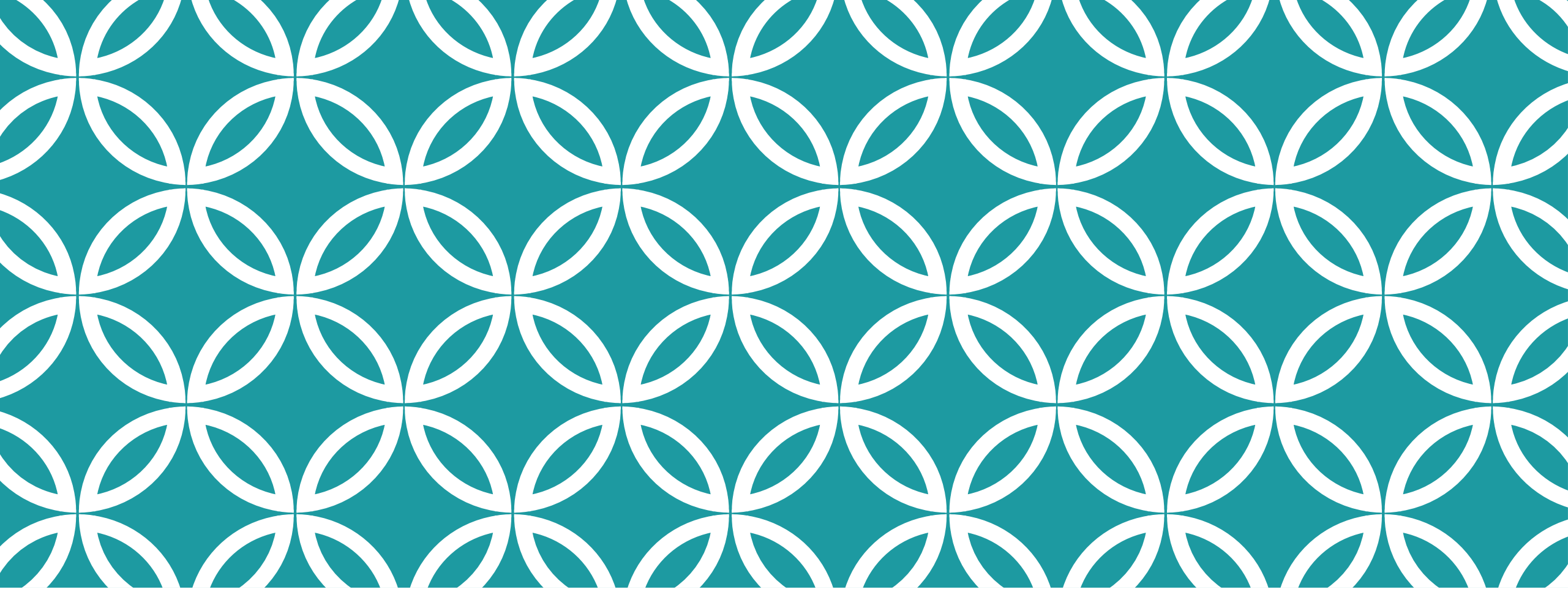




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-8: Whole numbers and Half Truths



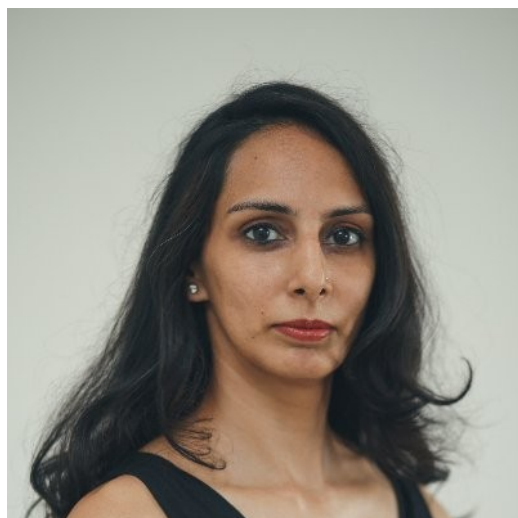


WHOLE NUMBERS AND HALF TRUTHS |

WHOLE NUMBERS AND HALF TRUTHS

What Data Can and Cannot
Tell Us About Modern India

RUKMINI S.



I HOW INDIA TANGLES WITH COPS AND COURTS

II WHAT INDIA THINKS, FEELS AND BELIEVES

III HOW INDIA (REALLY) VOTES

IV EAT, PRAY, ENJOY, LOVE, MARRY—HOW INDIA LIVES LIFE

V HOW MUCH MONEY DO INDIANS MAKE?

VI HOW INDIA SPENDS ITS MONEY

VII HOW INDIA WORKS

VIII HOW INDIA IS GROWING AND AGEING

IX HOW INDIA LIVES AND WHERE

X HOW INDIA FALLS SICK AND GETS BETTER

SOME REVIEWS

A few tidbits:

1. **Indian respondents expressed greater support for a 'strong leader' and for army rule than most other countries. The share of Indians who thought that a strong leader was 'very good' for the country was higher than in any other country—even Russia!**
2. **With 84 per cent of Indians saying that religion is very important in their lives, India is far more religious than Europe, Israel, Latin America and the United States.**
3. **Just nine countries around the world, including Syria and Iraq, have a smaller proportion of working women than India**, and if Bihar were a separate country, it would have the lowest share of working women in the world.
4. **India is actually no more than one-quarter to one-third vegetarian.**
5. Kerala eats the most fish, Jammu and Kashmir the most mutton, and Andhra Pradesh the most chicken.
6. **Marriage is nearly universal in India. By the age of forty-five-forty-nine, only 1 per cent of women and 2 per cent of men have never been married.**

Statistical Body	Data Function/Focus	Relevant Chapter Example
Census enumerators	Capture every single Indian.	Chapter 9.
National Statistical Office (NSO)	Survey households on consumption details (e.g., number of eggs eaten).	Chapter 6.
National Crime Records Bureau (NCRB)	Aggregates all First Information Reports (FIRs) from police stations.	Chapter 1.
National Family Health Survey (NFHS)	Discreetly asks women about experiences of assault.	Chapter 1.
Civil Registration System (CRS)	Captures births, deaths, and fertility rates.	Chapters 8 and 10.
Private Research (e.g., NCAER, CSDS, CMIE)	Check government data, conduct world-class opinion polls, provide high-frequency economic data, and capture insights on social attitudes.	Chapters 2, 3, 5, 7.

DATA – CRITICAL ANALYSIS

The problem is not a lack of data, but rather that inaccurate narratives are built without looking at the data, or because of "bad data".

Statistics alone are insufficient; they require **context, interpretation that is free from ideological spin, and critical engagement.**

Numbers as Enlightenment: Numbers are powerful tools that **can enlighten, empower, elevate, and help make sense of modern India and anticipate the future.**

The Political Context: Data does not exist in a vacuum; it is affected by political and social forces, leading to polarization in the world of data.

CHAPTER 1: HOW INDIA TANGLES WITH COPS AND COURTS

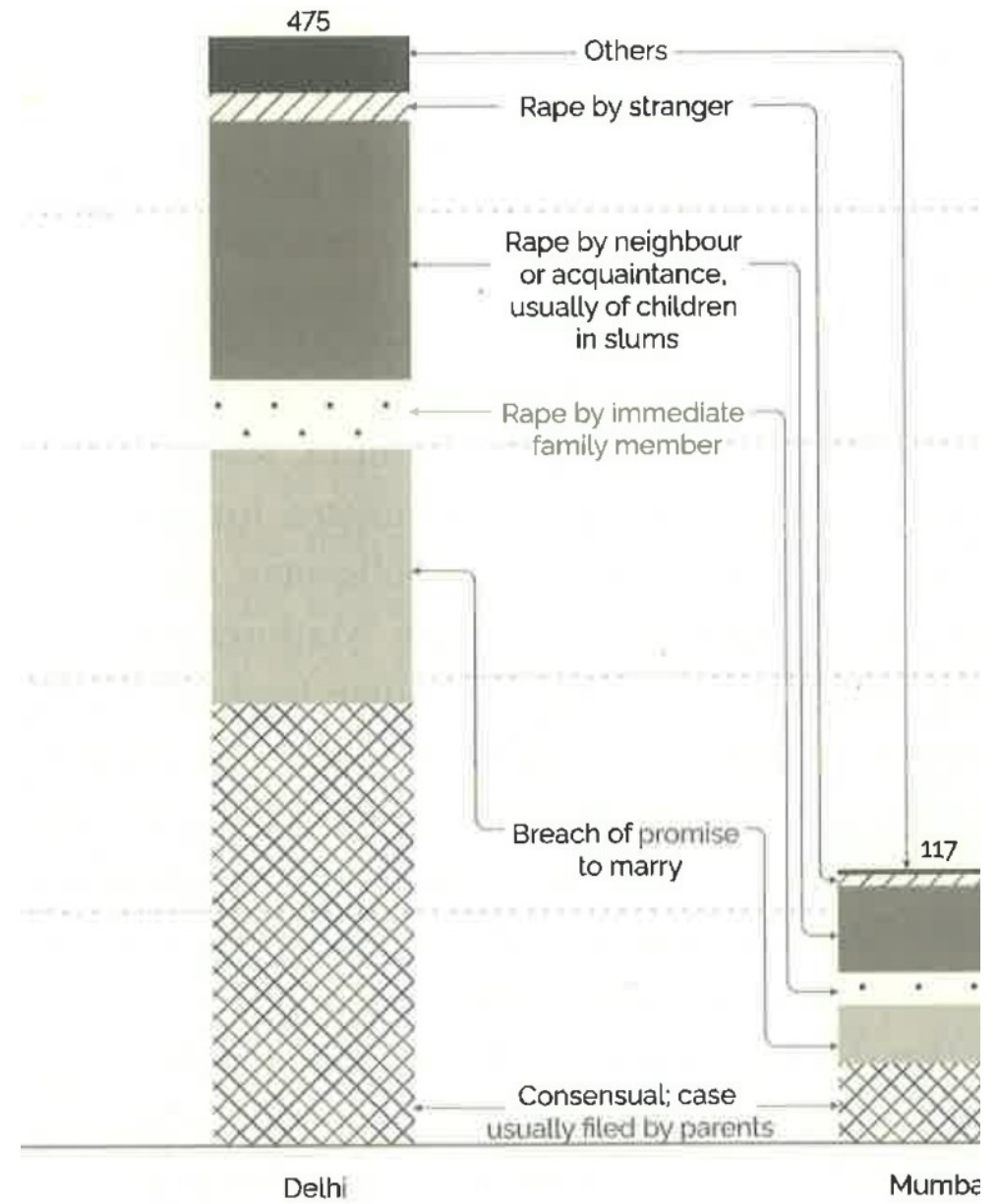
Relying solely on police statistics (primarily from the NCRB) and media reporting has created a **deeply distorted picture of crime in India**.

Flaws in Official Crime Reporting (NCRB)

The Principal Offence Rule (The Hidden Flaw): The NCRB adheres to the 'principal offence rule,' selecting **only the 'most heinous' crime** from each FIR for their national statistics.

(ex: **A case involving both sexual assault and murder would only be registered as murder because it carries the maximum penalty.**)

□ **Legal Framework Limitations:** The NCRB cannot produce data for crimes that lack a corresponding IPC section. For instance, data on mob lynchings and hate crimes was omitted from a 2019 report because the government stated that only existing murder sections were legally available tools, illustrating that the legal framework must precede data collection.



Delhi's Conviction Rate: **23%**

Mumbai's Conviction

MOVING CARS AND COLD DRINK BOTTLES

The Police Script: To ensure these cases make it to court (i.e., to prove lack of consent), a common informal script is followed: the victim is described as being abducted in a "**moving car**" or rendered unconscious by a "**sedative-laced cold drink**".

In a large analysis, **not one case produced proof of intoxication**; the "cold drink bottle was the smoking gun that never appeared".

Impact of Legal Reforms (Post-2012): The 2013 decision to raise the age of consent for sexual acts from sixteen to eighteen made it easier for families to add rape charges to kidnapping cases involving teenagers or young adults in relationships.

□ This ensures the male partner's arrest and separation of the couple. The law now requires a minimum sentence of ten years, effectively placing young men in consensual relationships with minors in the same category as paedophiles.

Breach of Promise to Marry: Cases involving alleged sexual consent obtained under a false promise of marriage are common, often filed when a man calls off a long-term relationship.

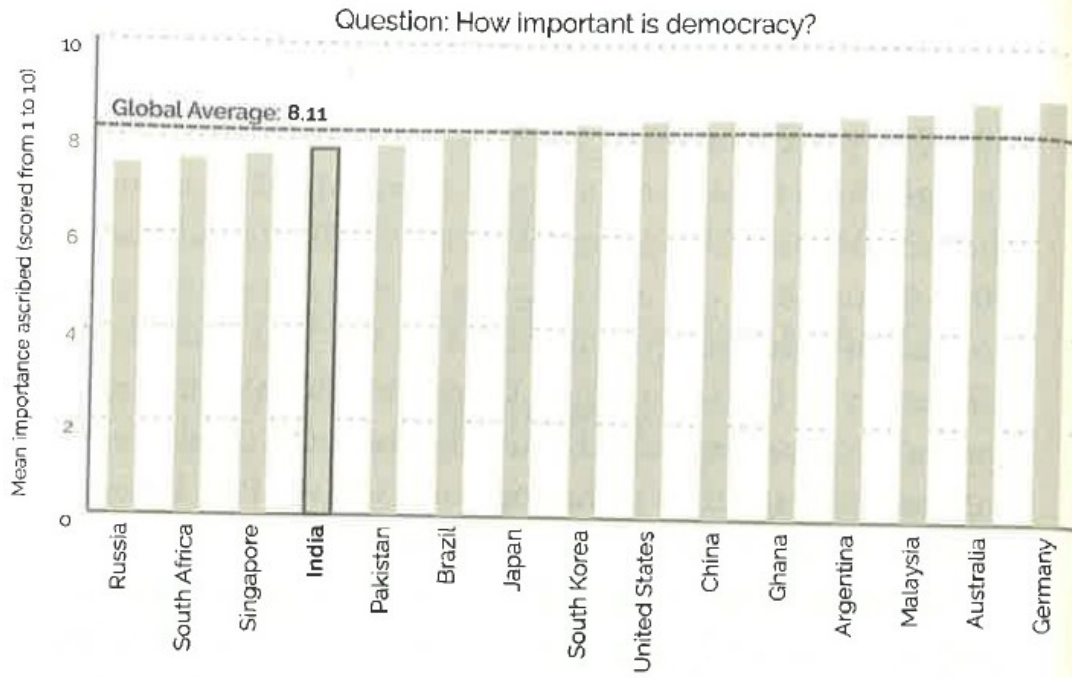
Low Conviction Rates and Misapplied Laws: While activists cite the low national rape conviction rate (27% in 2018) to highlight systemic bias, judges

Figure 1.2: Property crimes are least likely to make it to police records

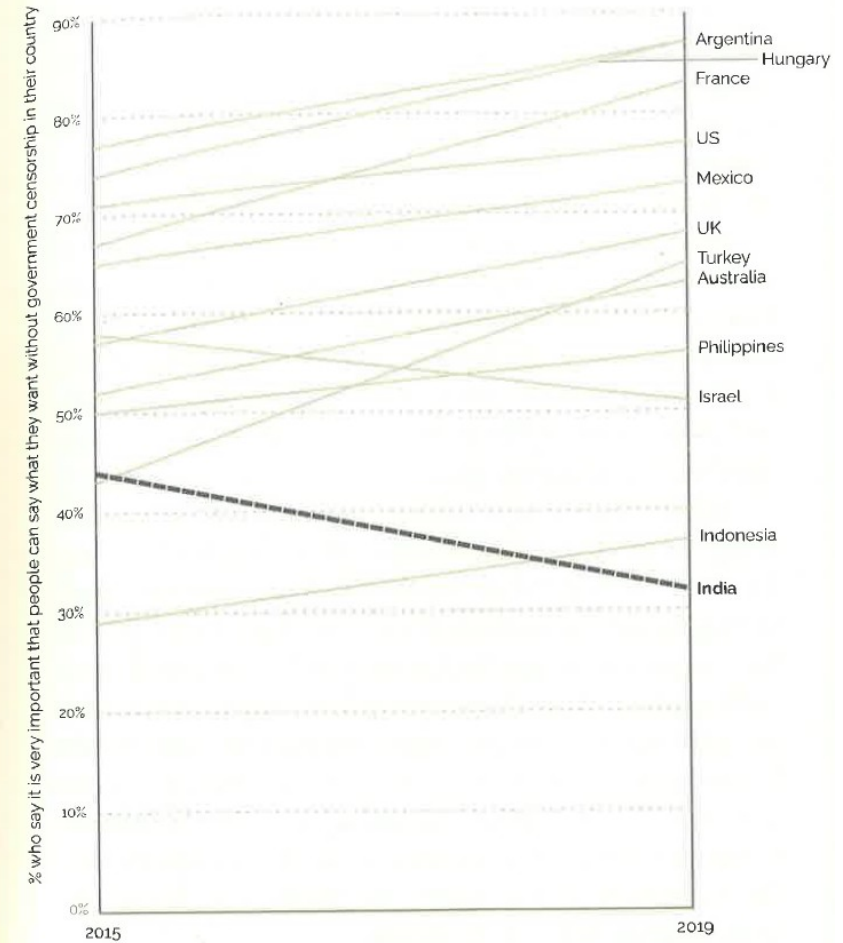


DEMOCRACY

India placed lower emphasis on democracy than most other countries



What's worse, India may be getting even less democratic



ILLIBERAL

Younger people do not have much more progressive beliefs; a 2017 survey on the attitudes of young people found that six out of ten respondents supported banning movies which hurt religious sentiments, even more so among Muslim youth, 70 per cent of Hindu youth were opposed to allowing anyone to eat beef, and one-third of young people opposed inter-caste marriage.¹⁸ (Fig. 2.2)

Muslim youth were much more likely than others to report having experienced religion-based discrimination. About one in every seven or 13 per cent of them said they had been discriminated against based on their Muslim identity. Muslim youth living in smaller cities were most likely to have been victims of religious bias—27 per cent reported having faced discriminatory treatment for being Muslim. This is double of what all Muslim youth reported.¹⁹

Figure 2.2: Younger people do not necessarily hold more liberal views

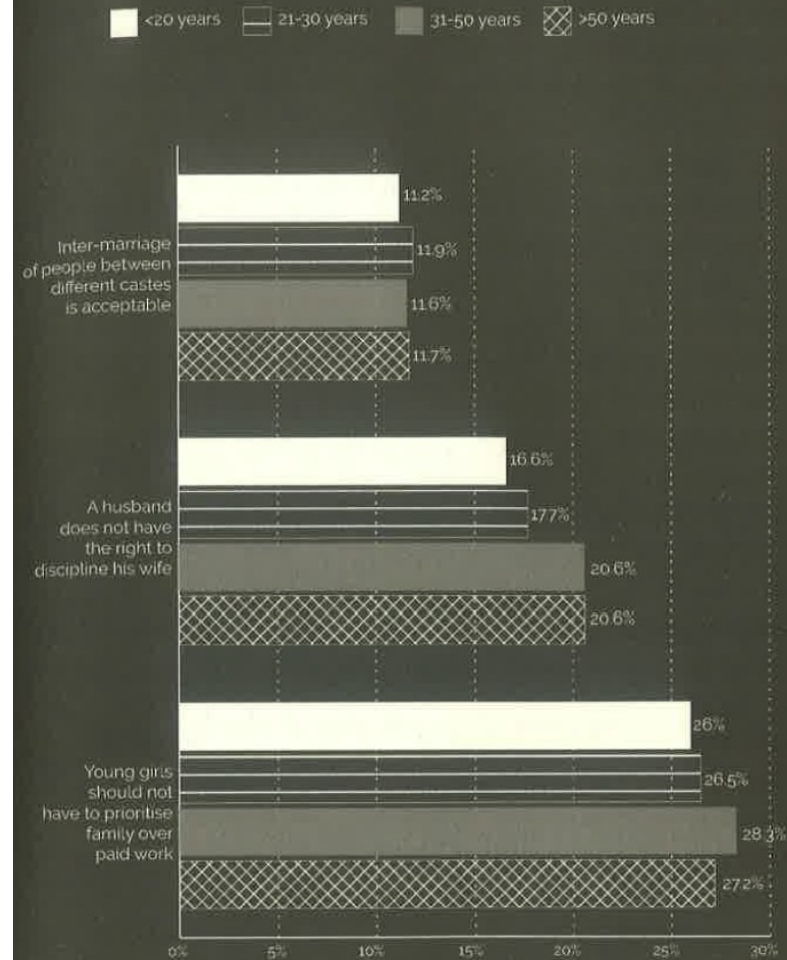
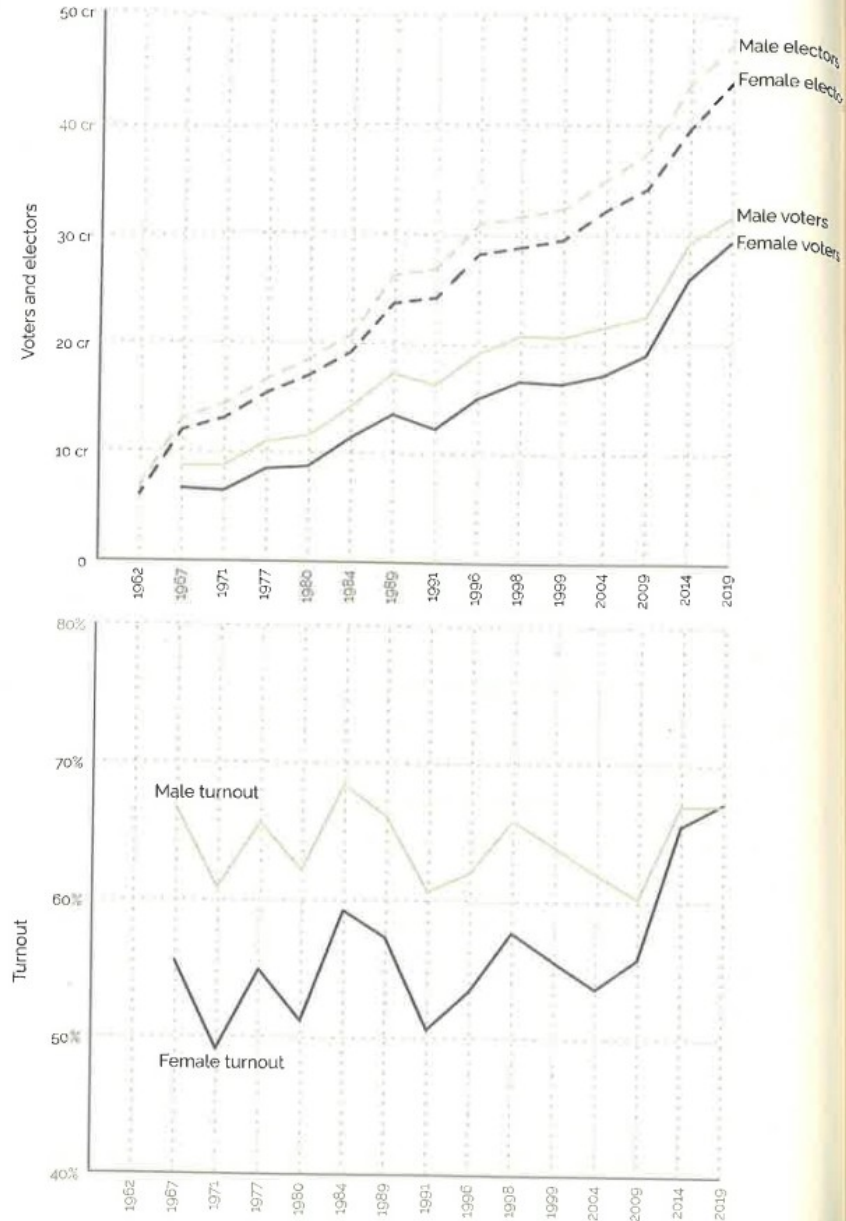


Figure 3.1: The increase in voter turnout in India is almost exclusively a female story



HOW INDIA VOTES

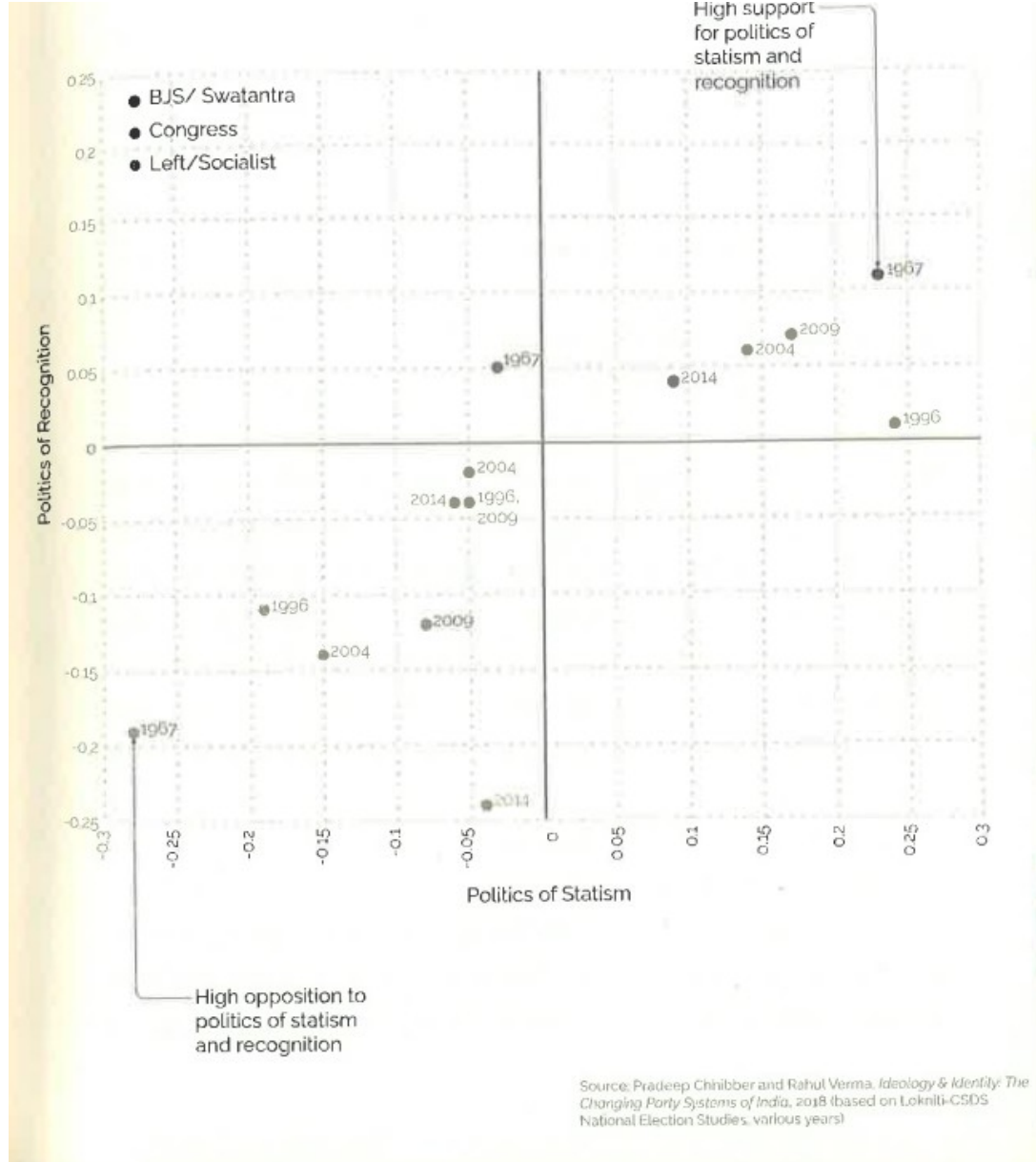
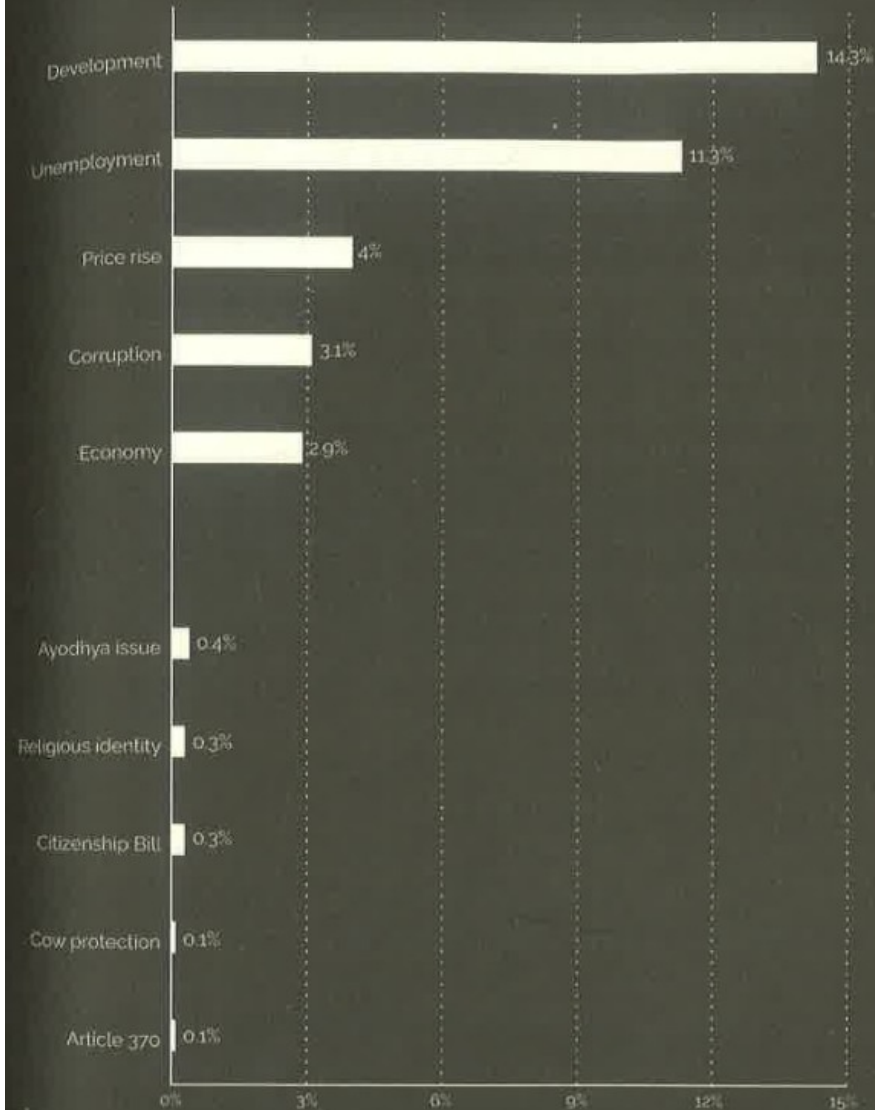
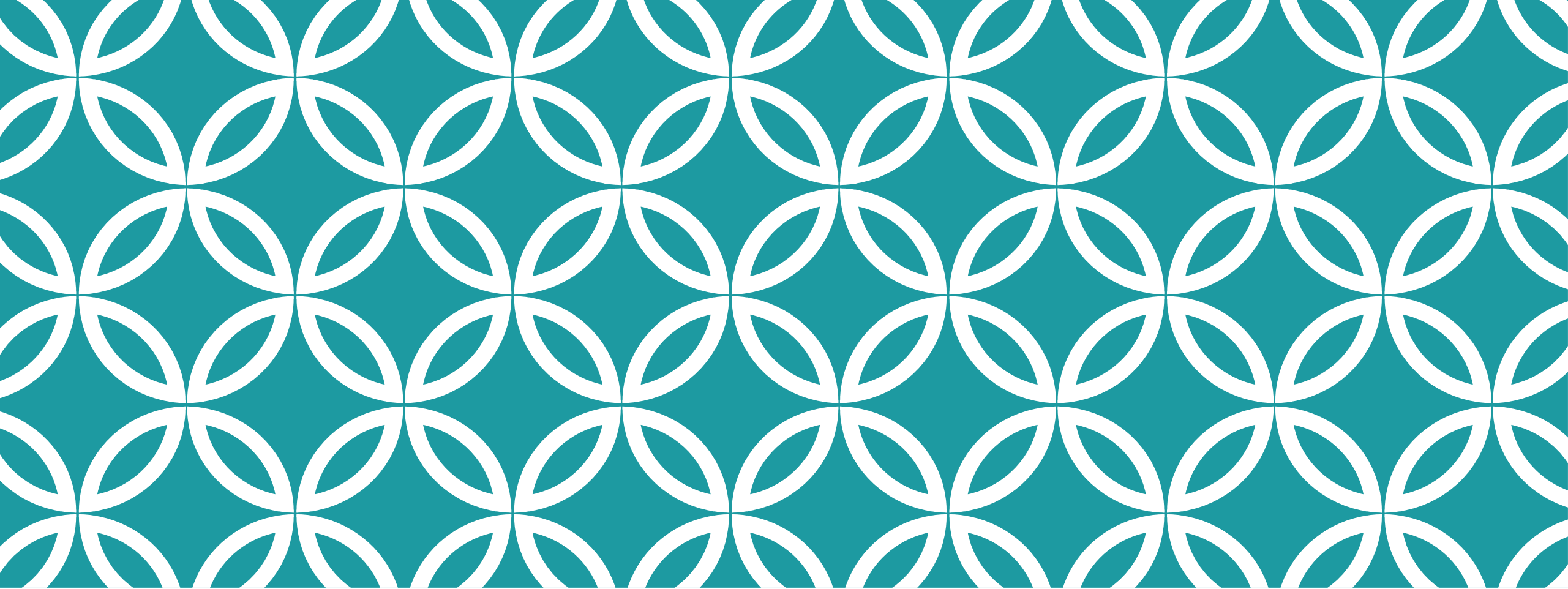


Figure 3.3: All voters claim they want development

What was the most important issue for you at the time of voting?





STUDENT DATA VISUALIZATIONS |

BEST VISUALS

Women

Men

Watches television at least once a week

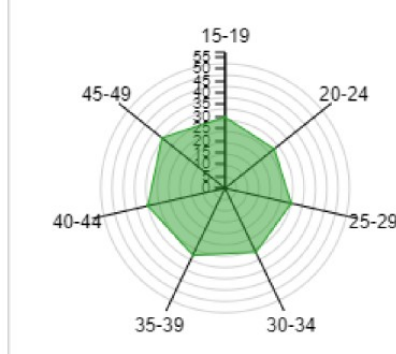
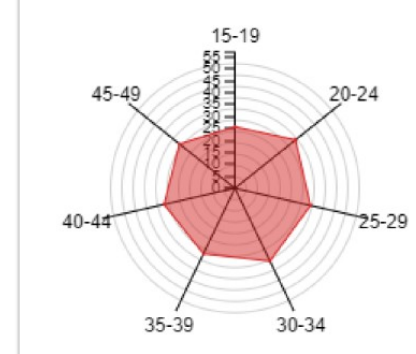
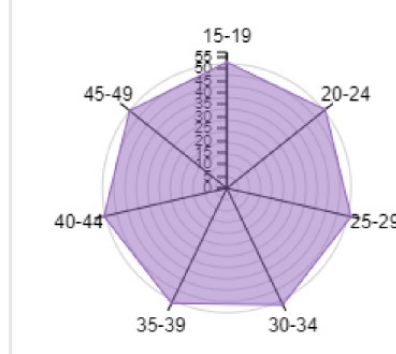
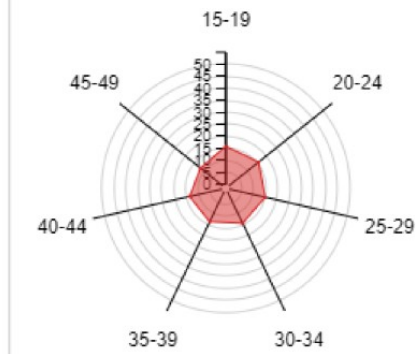
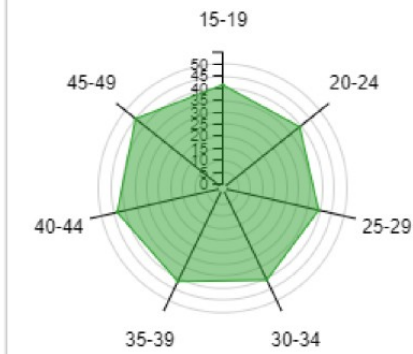
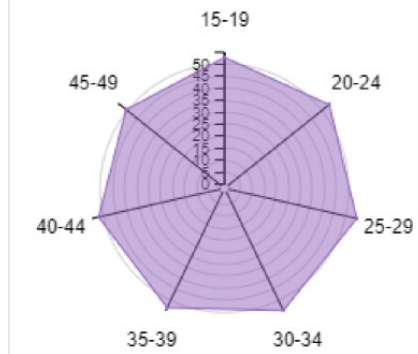
Not regularly exposed to any media

Reads Newspaper or magazine at least once a

Watches television at least once a week

Reads Newspaper or magazine at least once a

Not regularly exposed to any media

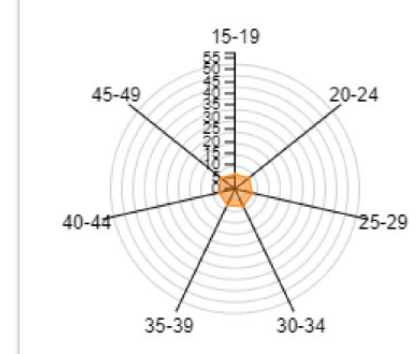
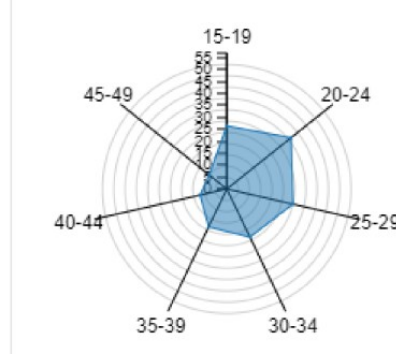
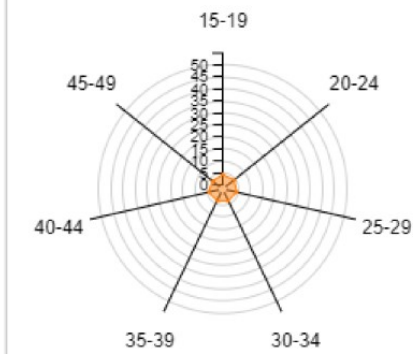
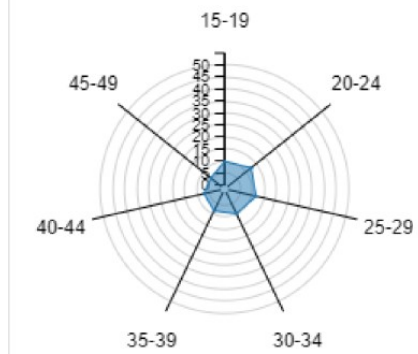


Visits the cinema/theatre at least once a month

Listens to the Radio at least once a week

Visits the cinema/theatre at least once a month

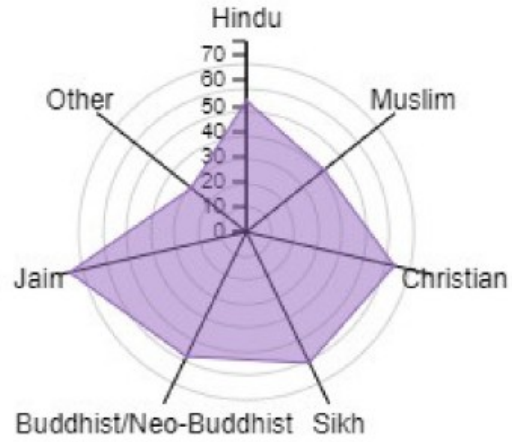
Listens to the Radio at least once a week



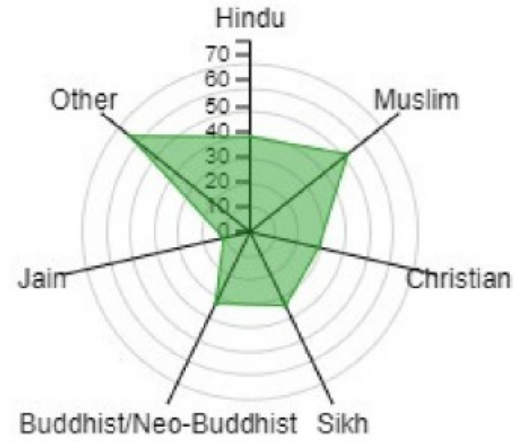
EXPOSURE TO MASS MEDIA

Women

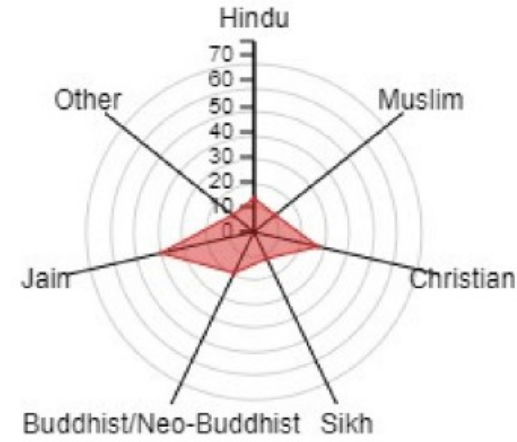
Watches television at least once a week



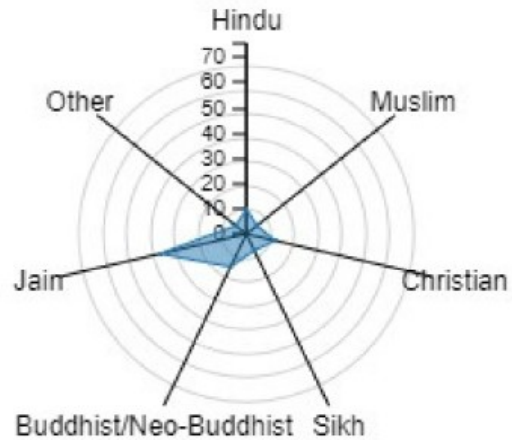
Not regularly exposed to any media



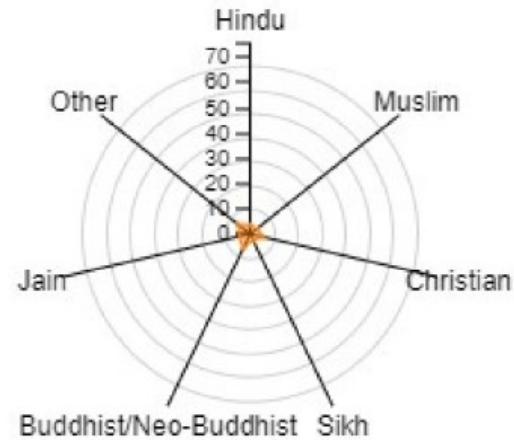
Reads Newspaper or magazine at least once a



Visits the cinema/theatre at least once a month



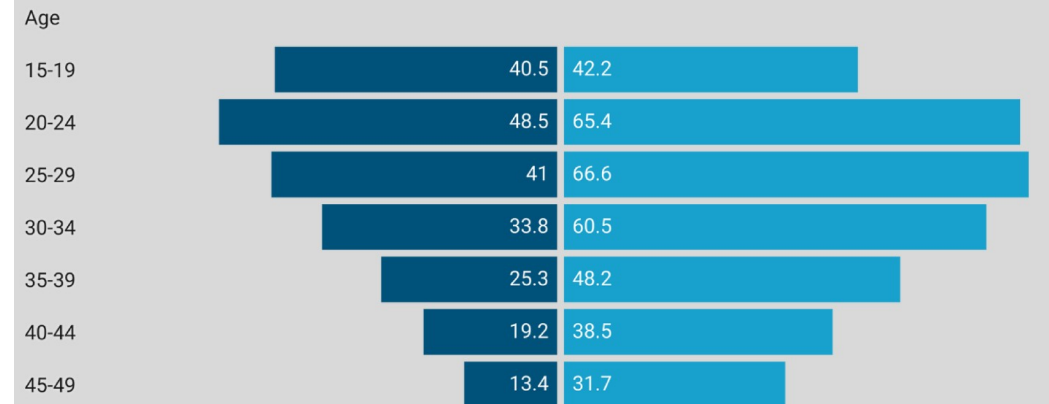
Listens to the Radio at least once a week



Internet Usage: (Age)

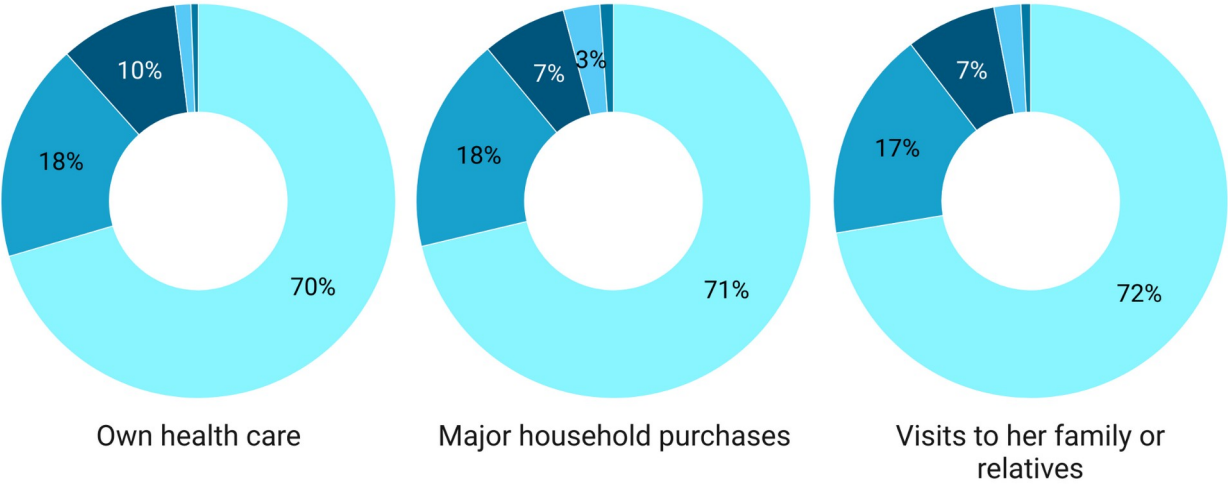
Women Men

Ever used Internet
Women Men



Participation in decision-making - Rural Women

Respondent and spouse jointly Mainly spouse Mainly respondent Someone else Other

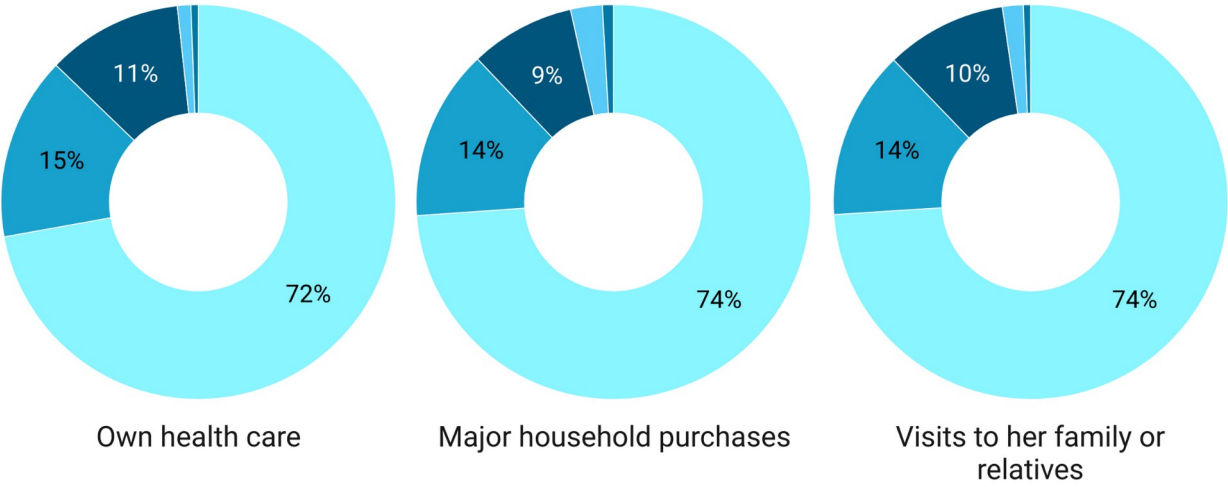


The majority of the decision making was done by the man and spouse jointly (average of 72%)

Mainly the spouses made decisions for rural women (at 18%) when compared to spouses of urban women. Also, the women residing in urban areas have minute but noticeable decision-making power than women residing in rural areas. This shows that urban women have more decision making power than rural women.

Participation in decision-making - Urban Women

Respondent and spouse jointly Mainly spouse Mainly respondent Someone else Other

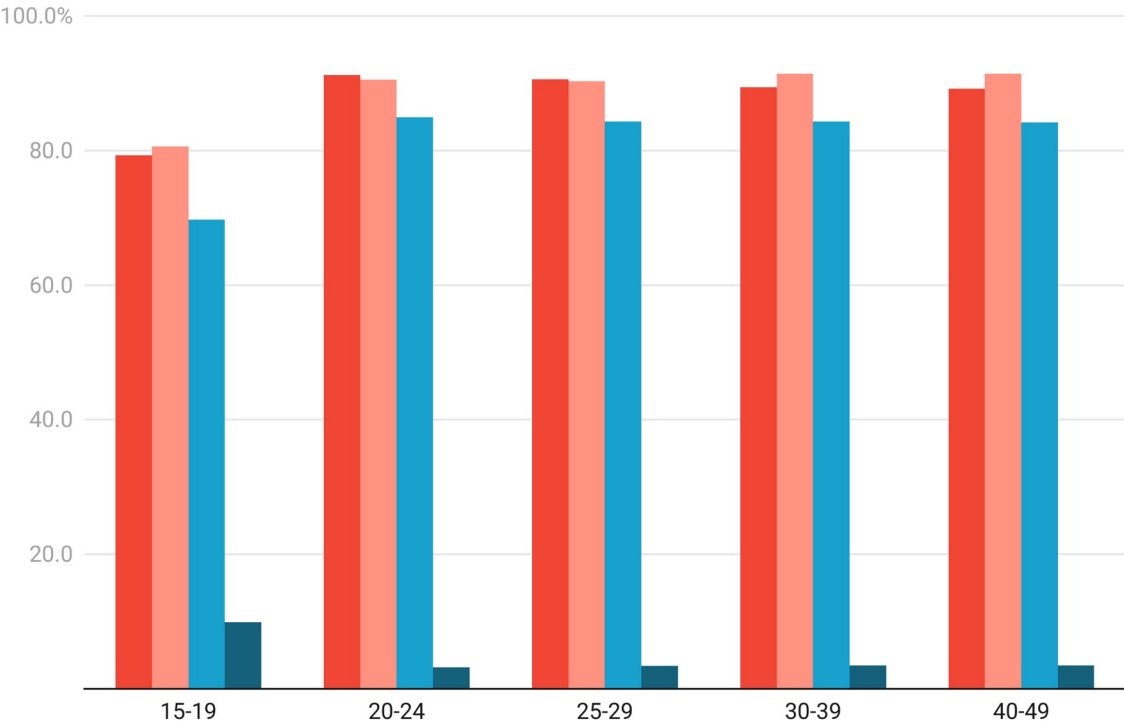


Both Urban and Rural women and their spouses jointly made slightly lesser decisions relating to own health care (at 71%), when compared to decisions regarding visits to family and relatives (73%).

Men's participation in decision making by Age

Percentage of men who usually make specific decisions alone or jointly with their wife (in shades of red)

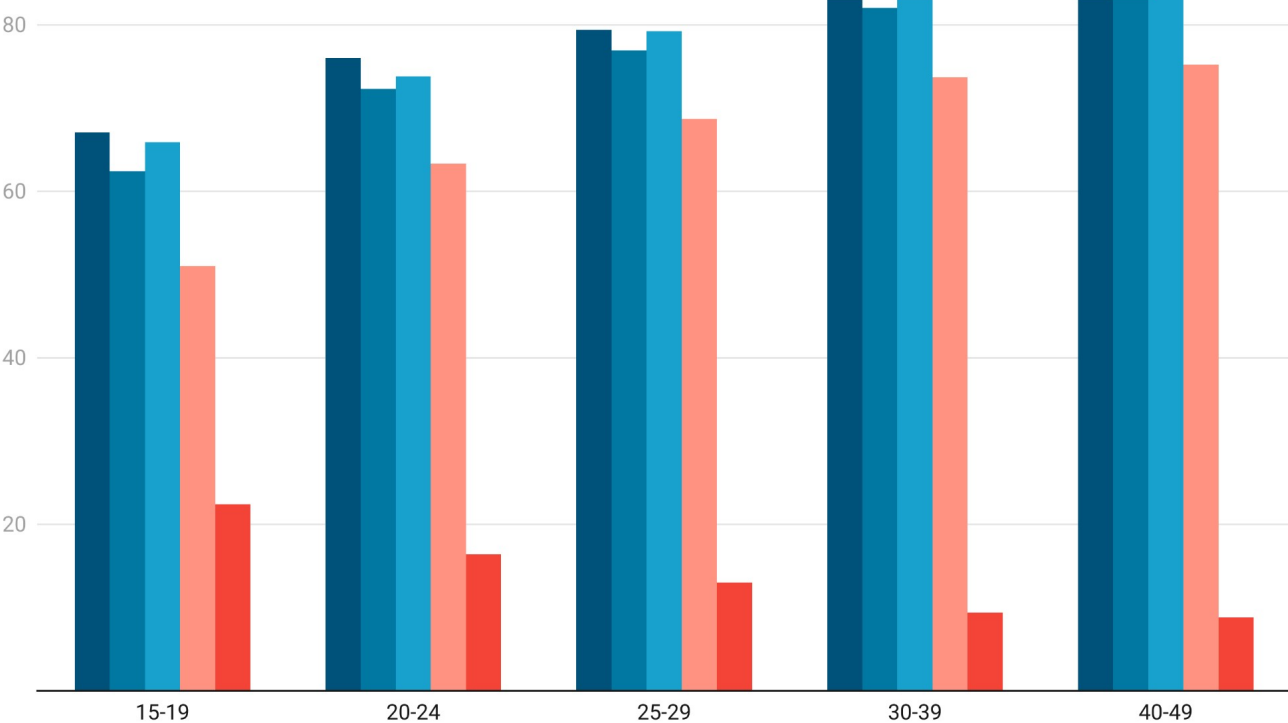
- Own health care
- Making major household purchases
- Percentage of men who participate in both decisions
- Percentage of men who participate in neither of the two decisions



Women's Participation in decision-making by Age

Percentage of women who usually make specific decisions alone or jointly with their husband (in shades of blue)

- Own health care
- Making major household purchases
- Visits to her family or relatives
- Percentage of women who participate in all three decisions
- Percentage of Women who participate in none of the three decisions

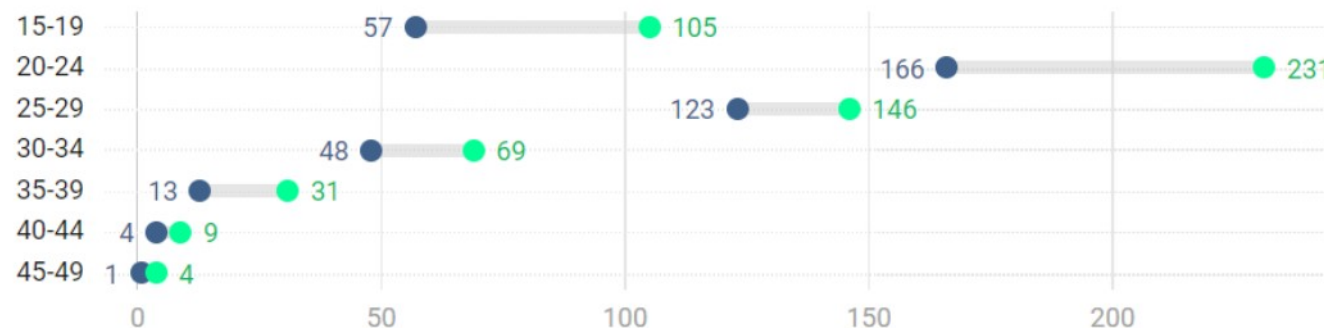


When it comes to decision making by age, for men, till the age of 19 since most of them are still minors. But between the age of 20-49 it does not seem like the age factor affects their participation. However, for women, we can see a direct correlation between participation and age; there is a distinct increase in overall participation with increasing age. This shows that while women gain more decision making power as they grow older, men's decision making power stays consistent throughout adulthood. The clear decrease in percentage of women who participate in none of the decisions further confirms this. Additionally, percentage of men who participate in neither of the 2 decisions between the age of 15-19 is significantly higher than that from ages 20-49.

Fertility based on Age (Table 1)

Urban VS Rural- Based on Age (NHFS 3)

The blue color dot indicates the Urban data and the green color dot indicates the Rural data



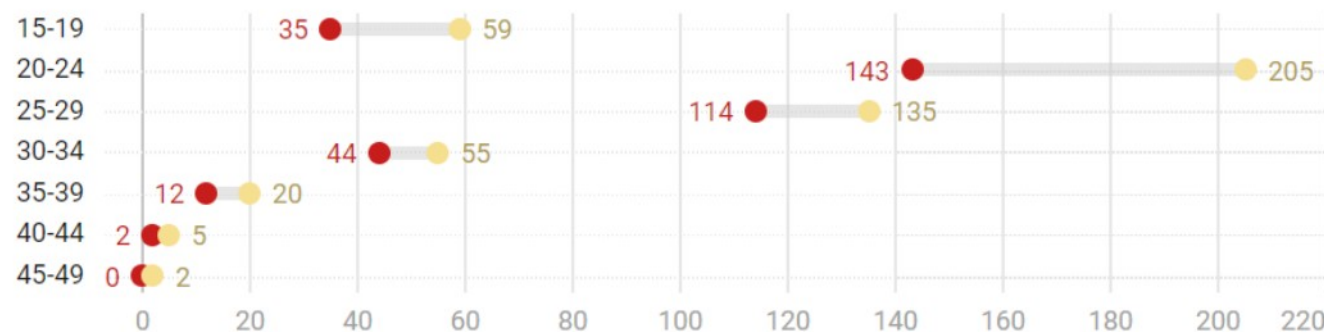
This chart shows the fertility rate of women in the urban and rural setting based on their age, recorded in NHFS 3

Chart: Sanjana Shah, Maahi Patel, Shrinidhi Suneeth • Source: [Ministry of Health and Family Welfare](#) • [Get the data](#) • Created with [Datawrapper](#)

The data given is pretty much what we thought it would be. The women in Rural areas used to (and in some families and states) get married pretty early on in life and start having children at an earlier age as well hence the large difference in the fertility rate. Women tend to have their children when they reach the age of 20-29 which is also evident from the chart. Towards the later years, that is 35 onwards the possibility of women getting pregnant decreases which is also depicted clearly in the chart.

Urban VS Rural- Based on Age (NHFS 4)

The red dot indicates the Urban fertility rate and the yellow dot indicates the Rural fertility rate



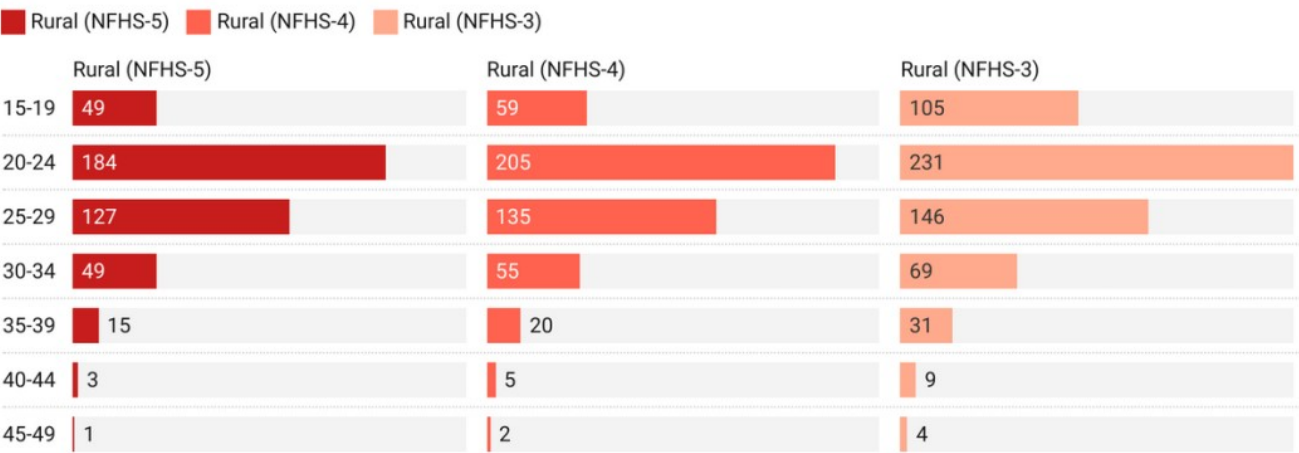
This chart shows the fertility rate of women in the urban and rural setting based on their age, recorded in NHFS 4

Chart: Sanjana Shah, Maahi Patel, Shrinidhi Suneeth • Source: [Ministry of Health and Family Welfare](#) • [Get the data](#) • Created with [Datawrapper](#)

A drastic change is seen in this particular chart. The number of pregnancies in all age groups have come down in both Urban and Rural settings. The difference between the Urban and Rural also has also reduced in comparison to the data from NHFS 3. The flow of the data in this chart remains the same in the sense that maximum pregnancies were between the age of 20 and 29 and the pregnancies began decreasing from the age of 35.

Current Rural Fertility

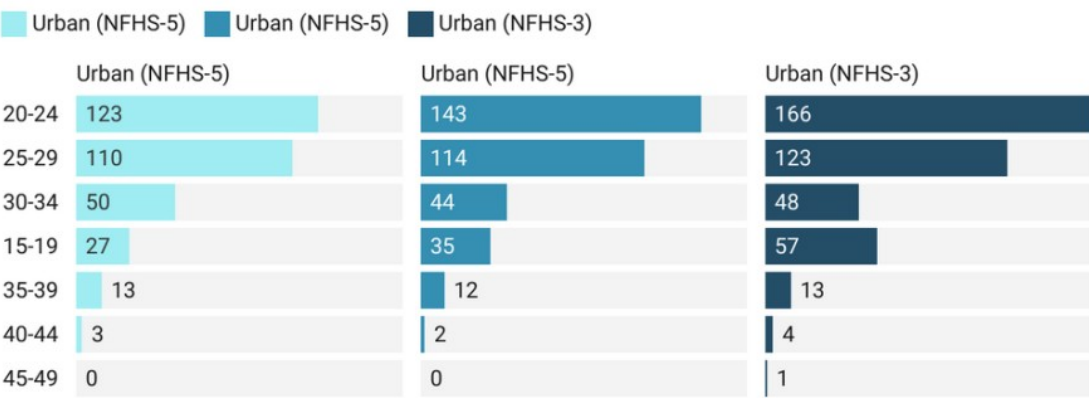
Age-specific Data



This data shows the age-specific current rural fertility rates from NFHS-5, NFHS=4, and NFHS=3.
Chart: Maahi Patel, Sanjana Shah, Shrinidhi Suneeth • Source: Ministry of Health and Family Welfare • Created with Datawrapper

Age specific current Urban fertility rates in India

This chart shows the increase or decrease in fertility rates through the surveys of Urban (NFHS-5), Urban (NFHS-4) and Urban (NFHS-3).

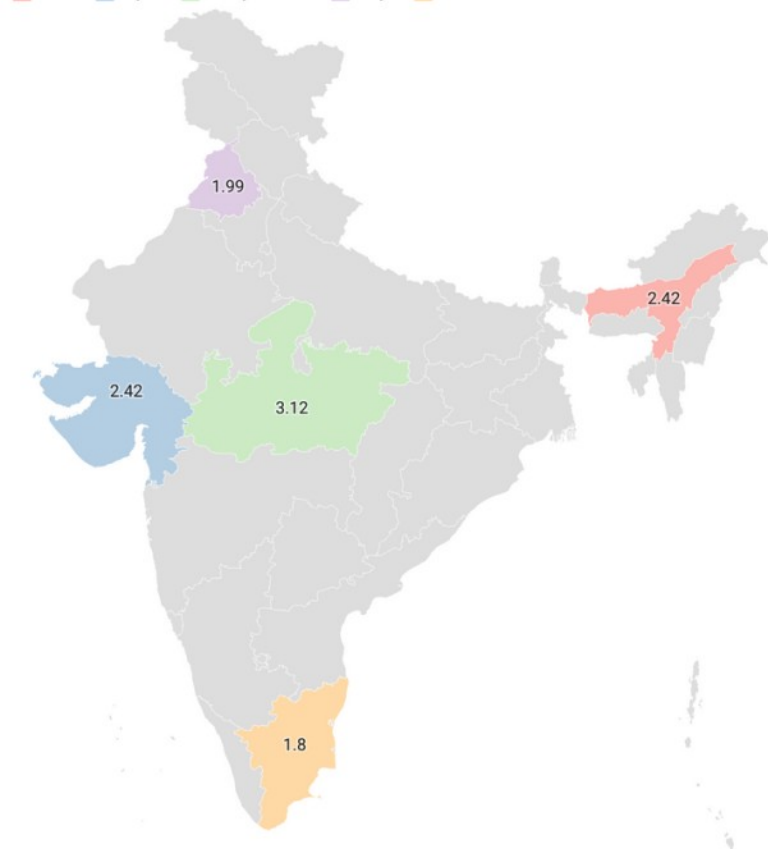


NFHS-3 (2003-05), NFHS-4 (2013-15), NFHS-5 (2017-19)
Chart: Maahi Patel, Sanjana Shah, Shrinidhi Suneeth • Source: Government of India, Ministry of Health and Family Welfare, National Family Health Survey (NFHS-5, 2019-21). • Created with Datawrapper

Total Fertility Rate of India, NFHS-3

This map shows the Total Fertility Rates of the selected states from all regions of India according to the NFHS-3 survey.

Assam Gujarat Madhya Pradesh Punjab Tamil Nadu

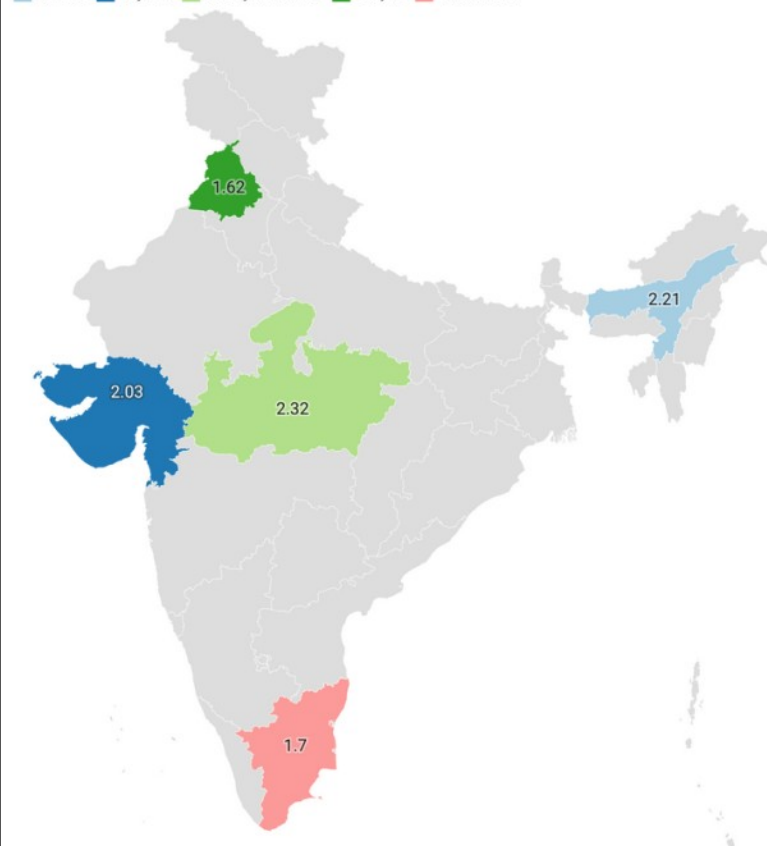


Map: Sanjana Shah, Shrinidhi Suneeth, Maahi Patel • Source: Government of India, Ministry of Health and Family Welfare, National Family Health Survey (NFHS-3, 2019-21). • Map data: © OSM • Created with Datawrapper

Total Fertility Rate of India, NFHS-4

This map shows the Total Fertility Rates of the selected states from all regions of India according to the NFHS-4 survey.

Assam Gujarat Madhya Pradesh Punjab Tamil Nadu

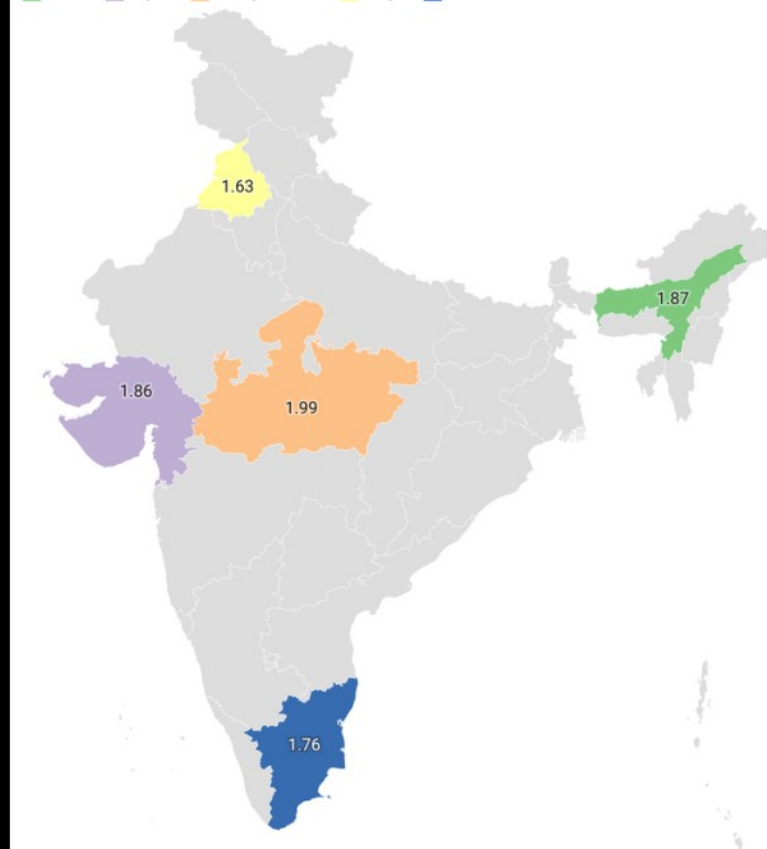


Map: Sanjana Shah, Shrinidhi Suneeth, Maahi Patel • Source: Government of India, Ministry of Health and Family Welfare, National Family Health Survey (NFHS-4, 2019-21). • Map data: © OSM • Created with Datawrapper

Total Fertility Rate of India, NFHS-5

This map shows the Total Fertility Rates of the selected states from all regions of India according to the NFHS-5 survey.

Assam Gujarat Madhya Pradesh Punjab Tamil Nadu



Map: Sanjana Shah, Shrinidhi Suneeth, Maahi Patel • Source: Government of India, Ministry of Health and Family Welfare, National Family Health Survey (NFHS-5, 2019-21). • Map data: © OSM • Created with Datawrapper